

Brandon Oltarsh

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Purdue University — Class of 2028

Major: Aerospace Engineering | **Minor:** Mathematics

GPA: 3.98/4.0

Portfolio: <https://brandonprog.github.io/brandonsportfolio/> (see past projects here)

Experience

Mechanical Director — FRC Robotics Team 2383

Jan 2022 – Apr 2024

- Led workshops and design sessions to train 40+ underclassmen in mechanical design, robot assembly, and workplace maintenance.
- Maintained robotics lab and machine shop operations supporting 60+ active members.
- Designed and built competition-ready robots, contributing to multiple award-winning competition and outreach events.
- Collaborated with peers to prepare and present for team recognition and community awards.

Senior Counselor & Design Instructor — Robotics Summer Camp

Summers 2022 – 2024

- Taught 50+ students (ages 10–14) mechanical design principles and CAD using Onshape.
- Created hands-on challenges, lectures, and robot base designs to reinforce engineering concepts.
- Coordinated daily logistics for 100+ campers across multiple programs, ensuring safety and engagement.

Cooling Analysis & Engine Optimization Member — Purdue Space Program (Active Controls)

2024 – 2025

- Assisted in developing MATLAB-based regenerative cooling analysis for a 500-lbf liquid rocket engine.
- Designed methodology to evaluate film cooling efficiency and material performance.
- Researched and compared alcohol-based propellants and metals to optimize efficiency and thermal management.

Volunteer — Craig's Pantry

Aug 2018 – May 2024

- Packaged and delivered food and clothing to 200+ food-insecure families in Broward County.
- Organized large community donation drives, securing hundreds of pounds of food and clothing.

Relevant Course Work: Thermodynamics, Aeromechanics, Physics Mechanics, NX PDM

Skills-

Engineering Tools: NX, Siemens PDM, Onshape, Autodesk Inventor, CAM, CNC Machining (100+ hrs), FDM 3D Printing

Programming: MATLAB, Python, HTML, CSS, C

Productivity: Excel, Word, PowerPoint, Google and Microsoft Suite

